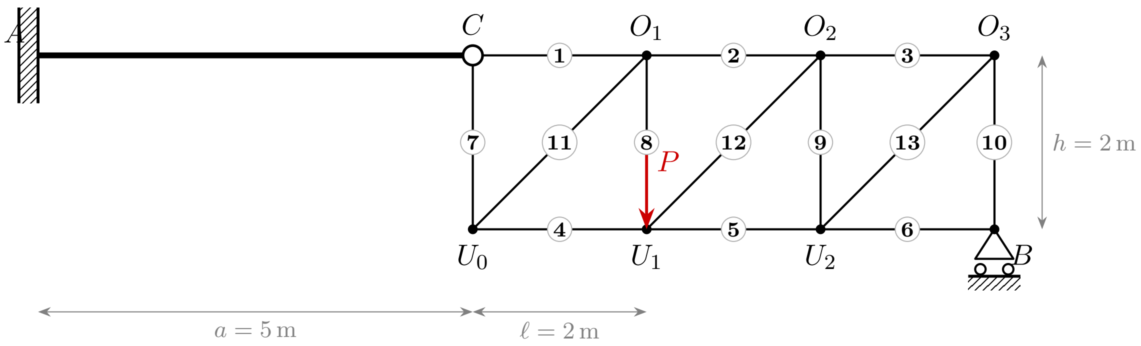
University of Natural Resources and Life Sciences, Vienna (BOKU)

VU Mechanik – LAWI 100515 (PI)

Exam **Group A** Date: 26.06.2026

Model solution



1. Static determinacy

Counting criterion $n = r - (3 + v)$ with r support reactions and v hinge conditions:

$$n = r - (3 + v) = 4 - (3 + 1) = 0 \quad \Rightarrow \quad \text{the system is statically determinate.}$$

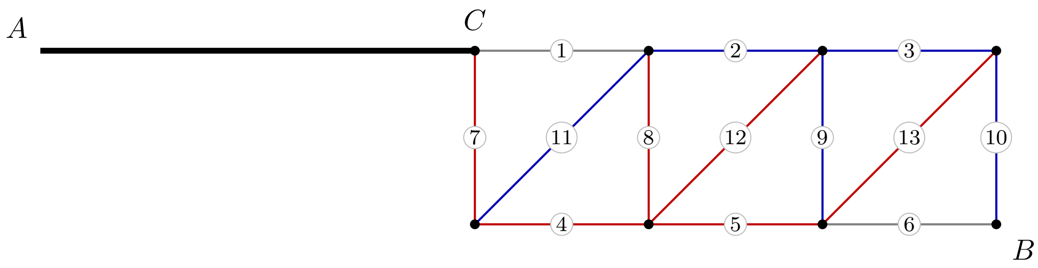
2. Support reactions and hinge force

$A_x = 0\,\mathrm{kN}$, $A_y = 8\,\mathrm{kN}$, $M_A = 40\,\mathrm{kNm}$, $B_y = 4\,\mathrm{kN}$
$C_x = 0\,\mathrm{kN}$, $C_y = -8\,\mathrm{kN}$

3. Truss member forces

Member	Member force S [kN]	Type
1 (C – O1)	0	zero-force
2 (O1 – O2)	-8	compression
3 (O2 – O3)	-4	compression
4 (U0 – U1)	8	tension
5 (U1 – U2)	4	tension
6 (U2 – U3)	0	zero-force
7 (C – U0)	8	tension
8 (O1 – U1)	8	tension
9 (O2 – U2)	-4	compression
10 (O3 – U3)	-4	compression
11 (U0 – O1)	-11.31	compression
12 (U1 – O2)	5.66	tension
13 (U2 – O3)	5.66	tension

red: tension (+) blue: compression (−)



4. Section-force diagrams of the bending beam A–C